

INJURY PATTERN IN ELITE INDIAN MEN FIELD HOCKEY PLAYERS - PRE OLYMPIC 2012

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ABSTRACT

The purpose of this epidemiological study was to provide a descriptive account of injuries of Indian elite men field hockey players who participated in the preparatory training camp for London Olympic 2012. The sample of this study consisted of 82 male players- 30 senior players (Age range-17-32 years) and 52 junior players (Age range- 16-29 years) - who participated in the Indian men hockey team training camp held at Sports Authority of India, Bangaluru Centre during 2011-12. The data were collected by a qualified physiotherapist using pre-structured questionnaire and clinical examination. 73.2% players sustained a total of 68 injuries. Injury to lower limb was most common followed by upper limb and torso, but there was no injury to head, neck and face. Strain to hamstring, quadriceps and adductor muscle of thigh and lateral ligament sprain of ankle were the most commonly encountered intrinsic injuries whereas contusion to anterior thigh, shin, foot, wrist and hand was the most common extrinsic injury. Junior players mostly sustained contact injuries whereas the senior players sustained strain.

KEY WORDS

Injury pattern, epidemiological study, field hockey, injury prevention

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INTRODUCTION

Though one of the most popular team sports in the world; Field Hockey is an inherently a dangerous contact sport, contact within the game can occur with the ball, stick, another player or the playing surface. Injuries of musculoskeletal system are the inherent risk of sport participation (Engebretsen & Bahr, 2009). The physical strain of hockey play has been shown to be considerable, there is a greater injury risk inherent in playing on synthetic surfaces than on grass (Thomas Reilly, Andrew Borrie, 1992). Injuries are the one of the main reasons for inconsistent performance and premature termination of sporting career of many players. In the recent years protection of the health of athletes has become one of the priority areas of IOC and its Medical Commission is emphasizing on developing injury prevention protocol in all the sports so that sports participation remains risk free. Emphasis on the prevention of injuries during competition and training has the potential to bring about significant reduction in the incidence of sports injuries that ultimately enhances the sports performance (Engebretsen & Bahr, 2009). National Collegiate Athletic Association (NCAA) has developed a mechanism on injury information so that necessary evidence

based decision making with regard to health and safety policy can be made (Randall Dick et al., 2007). Hockey is a sport where high rates of injuries have been reported worldwide (Rishiraj et al, 2009; Dick et al 2007). Severe form of injuries noticed in field hockey that include trauma to the head, face, and upper limb and usually are the result of contact with the stick or ball; looking into the future, research should focus on developing and Evaluating effective strategies for injury prevention (Murtaugh 2009). An understanding of epidemiology of injuries in a given sports is a prerequisite for effective prevention and management strategies. Unfortunately not much data are available on the epidemiology of injuries of Indian sports persons.

The aim of this study was to provide a descriptive account of injuries of Indian elite men field hockey players who participated in the preparatory training camp for London Olympic 2012.

MATERIAL AND METHOD

The sample of this study consisted of 82 male players out of which 30 were senior players (age range-17-32 years) and 52 junior players (age range- 16-29 years) who participated in the Indian men hockey team training camp held at Sports Authority

of India, Bangaluru Centre during 2011-12. The training was conducted on synthetic surface. The data were collected during 25th March to 10th April 2012 by a qualified physiotherapist using pre-structured questionnaire and clinical examination. The questionnaire focused on personal characteristics (age, height, and weight), mode of injury, type and state of injury, body part and tissue involved and the mechanism of injury. The severity and type of tissue involved in the injury was determined by clinical examination using standard clinical protocol. For the purpose of this study, an Injury was defined as any injury occurring during the practice session or game and causing the player to stop playing or to miss the next practice session or game. This definition was adapted after Tegner and Lorentzon (1991). Descriptive statistics was used to prepare summery statistics and are reported here in absolute

numbers and percentage.

RESULT

The demographic data of Indian elite junior and senior field hockey players is presented in Table- I. Out of 82, sixty players (73.2%) sustained a total of 68 injuries. Among these injured players 8 (senior 5, junior 3) players sustained more than one injury. 22 out of 82 players did not report any injury. 38.2% injuries were sustained by the senior players, while the junior players sustained remaining 61.8% of injuries.

Table- II presents the data of mechanism of injuries among senior and junior players. Of all the injuries, 29 (42.6%) were contact injury while 39 (57.4%) were non-contact injuries. The highest number of injuries among senior players resulted from non- contact (85.7%) injuries, followed by stick (9.5%) and ball

Table-I: Demographic details, Injury Rate among Senior and Junior Players

Variable	Mean	SD(Mode)	Range		Senior	Junior	Total
Age (Yrs)	19.98	3.37 (19)	16-32	Non Injured	9	13	22 (26.82%)
Height (cm)	175.12	6.38	160-192	Injured	21	39	60 (73.17%)
Weight (kg)	66.20	7.15	52-85	Total	30	52	82 (100%)

Table-II: Comparison of Mechanism of injury

	Senior	Junior	Total
Contact	3	26	29 (42.6%)
Non contact	23	16	39 (57.4%)
Total	26	42	68 (100%)

(4.8%) contact. Among the junior players, trauma from ball contact (38.5%) resulted in highest number of injuries, followed by non- contact (35.9%) injuries and fall without contact (15.4%) injuries.

The classification of injuries as per anatomical part is presented in Table- III. Overall, maximum numbers of injuries were confined to lower limb followed by upper limb and low back, no player

sustained any injury to head neck and face (Fig. 1). Among the senior players, the most frequently injured site of the body was ankle/foot (30.8%), followed by hip/thigh and knee/leg accounting for 26.9% injuries each, and whereas among the junior players, the knee/leg (28.6%) was most frequently injured site, followed by hip/thigh (21.4%) and ankle/foot and wrist/hand accounting for 14.3% injuries each.

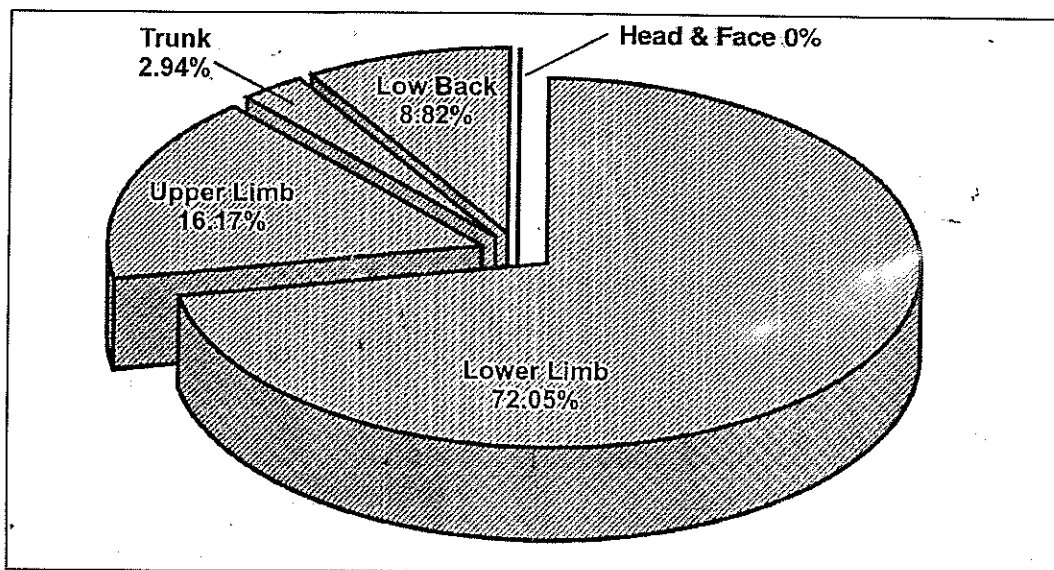
**Figure-1 : Distribution of injuries according to body segment**

Table-III: Distribution of injuries according to anatomical location

Location	Senior Players (n=30)	Junior Players (n=52)	Total
Head/Face/Neck	0	0	0 (0.0%)
Trunk	1 (3.84%)	1 (2.38%)	2 (2.94%)
Shoulder/ Arm	0	2 (4.76%)	2 (2.94%)
Elbow/Forearm	0	2 (4.76%)	2 (2.94%)
Wrist/ Hand	1 (3.84%)	6 (14.28%)	7 (10.29%)
Low Back	2 (7.69%)	4 (9.52%)	6 (8.82%)
Hip/ Thigh	7 (26.92%)	9 (21.42%)	16 (23.52%)
Knee/ Leg	7 (26.92%)	12 (28.57%)	19 (27.94%)
Ankle/ Foot	8 (30.76%)	6 (14.28%)	14 (20.58%)
TOTAL	26 (100%)	42 (100%)	68 (100%)

Table- IV presents the distribution of injuries according to type of injuries. Among the senior players, the most prevalent type of injury was strain (42.3%), followed by sprain (23.1%) and contusion (19.2%), while

contusion (42.9%) was the most prevalent type of injury among the junior players, and followed by strain (21.4%) and sprain (14.3%) (Fig 2). Strain to hamstring, quadriceps and adductor muscle of thigh and

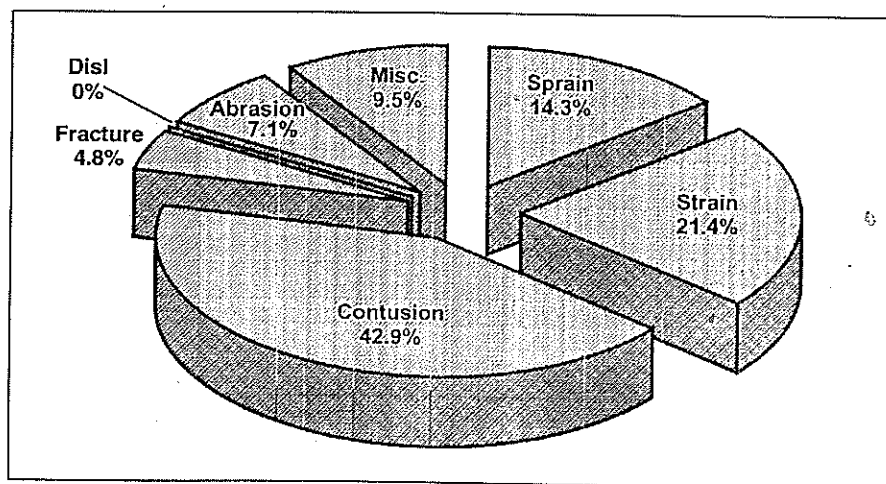


Figure-2 : Distribution of injuries according to tissue injuries

lateral ligament sprain of ankle were the most commonly encountered intrinsic injuries. We could observe only one injury to the medial ligament of knee. Contusion to anterior thigh,

shin, foot, wrist and hand was the most common extrinsic injury. Six players complained of prolonged shin pain aggravated by practice.

Table-IV: Distribution of injuries according to Type of injury

Type of injury	Senior Players (n=30)	Junior Players (n=52)	Total %
Sprain	6 (23.07%)	6 (14.28%)	12 (14.3%)
Strain	11 (42.30%)	9 (21.42%)	20 (21.4%)
Contusion	5 (19.23%)	18 (42.85%)	23 (42.9%)
Fracture	0	2 (4.76%)	2 (4.8%)
Dislocation	0	0	0 (0%)
Abrasion	0	3 (7.14%)	3 (7.1%)
Misc.	4 (15.38%)	4 (9.52%)	8 (9.5%)
TOTAL	26 (100%)	42 (100%)	68 (100%)

As for severity (Table- V), two (4.8%) injuries among senior and junior players, injuries were classified as major (Grade

III). Both these injuries were fractures of

fingers in junior players due to being struck

by ball. Majority of the injuries accounted

as Grade I, 24 (92.3%) and 40 (95.2%)

respectively.

DISCUSSION

The pattern of injuries observed in this study supports the observations of other

Table-V: Distribution of injuries according to Severity of injury

Severity	Senior	Junior	Total
Grade 1	24 (92.3%)	40 (95.23%)	64 (94.17%)
Grade 2	2 (7.69%)	-	2 (2.94%)
Grade 3	-	2 (4.76%)	2 (2.94%)
Total	26	42	68

authors albeit with some important differences. Injury to lower limb was most common followed by upper limb and torso, but we could not observe any injury to head neck and face.

Head and face injuries in field hockey are a concern and some studies have reported a high incidence of head neck and face injuries in hockey. According to Murtaugh (2001) the second most frequent site of injury in field hockey is the head and face region; increased exposure time and tendency to take more risks have been attributed as the factors increasing the likelihood of face injuries. Mukherjee (2012) documented total of 24 head and face injuries in men's field hockey Junior World Cup 2009 and reported the overall injury frequency rate of 16 per 1000 match hours and 19 per 1000 player matches. There were 92% of injuries that were of contact type, with the head being the most common site and contusion being the most common type of injury. Kaur et al (2008) in a survey on the Indian hockey players reported that abrasion to face, ocular trauma, dental injury and concussion comprised of 4.32% of total injuries. Kumar et al (2011) recorded a single incidence of cut injury at the corner of eye during penalty in the 15th Asian Games held at Doha in 2006. Dick et al (2007) reviewed 15 years of National Collegiate

Athletic Association (NCAA) injury surveillance data for women's field hockey and stated that concussion and head laceration injuries increased over this same time, and the risk of sustaining a concussion in a game was 6 times higher than the risk of sustaining one during practice. Most head/neck/face (71%) injury in their series occurred when the player was near the goal or within the 25-yd line and was caused by contact with the stick or ball. A reason for not sustaining any head or face injuries in this study could be the recording of injuries during practice session where it is plausible that players might not have played with full vigor or might have played cautiously.

The observation of this study supports the notion that lower limb is the most frequent site of injury in field hockey. Jamison & Lee (1989) reported that injuries to the lower extremities were the most commonly reported injuries among elite field hockey players playing on grass and synthetic surface. Fox (1981) suggests that the stooped position used when dribbling the ball may be an unsound position for fast locomotion and could contribute to lower limb injury. Kaur (2008) in a survey of Indian hockey players reported a high rate of occurrence of ankle sprain, shin pain and hamstring strain. Kumar et al (2011) reported that during Asian championship 71.6 % of injuries of Indian women hockey

team were confined to lower limb. They reported the highest (35.6%) occurrence of thigh and calf muscle related injury followed by ankle injury (20.0%) and knee injury (15.6%).

In the upper limb a hit from ball and fall to the ground contributes to most of the wrist and hand injuries. In the present study two players sustained fracture of thumb due to hockey stick hit. Out of three injuries sustained to the upper extremity proximal to wrist two were contusion due to ball hit and one was muscle strain due to fall. Recent surveys of collegiate and high school field hockey (including the current study) indicate that 14.0% to 15.8% of the total injuries are to the upper limb and most of these are wrist or finger fractures. According to 15 years injury surveillance data for women's field hockey of NCAA, 68% of hand/finger/thumb injuries occurred when the player was near the goal or within the 25-yd line and were caused by contact with the stick or ball (Dick et al 2007). In the survey of Kaur et al., 38.09% of injuries occurred to upper limb with wrist and hand being the most frequent site. Of participants in four stick-handling sports, collegiate field hockey players have significantly higher odds of sustaining hand or phalangeal injuries (Bowers AL et al., 2008).

Injuries in hockey can be classified into

two broad categories of contact and non contact injuries. Contact injuries are also referred in literature as extrinsic injuries and result from being struck by a ball or stick or by collision with other player goal post or ground. The finding of this study that contusion are the most common contact injuries in hockey is in agreement with other studies (Sherker & Cassell, 2002).

A difference was observed in the types of injuries sustained by junior and senior players. While the number of ligament sprain was equal for both levels of players the incidence of contusion was significantly higher in junior players whereas the senior players mostly encountered muscle strain. The contusion with ball and stick may imply the skill deficiency in the players. It seems that due to better technique and superior playing skills the senior players have avoided being hit by ball or stick. Murtaugh (2001) has suggested that players need to develop basic skills before progressing to a more competitive level of play.

CONCLUSION

This descriptive epidemiological study demonstrated that most prevalent training injuries in field hockey are lower limb injuries comprising of ankle sprain, strain of thigh muscle and shin splint followed by hand and finger injuries. It also revealed the difference in the pattern and mechanism of injury

between senior and junior level elite players. Injuries by ball and stick are. Even though most significant differences in injury rates, types of injuries were not severe. It does call for of injuries and mechanism of injuries were attention to develop better technical skill found among senior and junior Indian men among junior level players and emphasis on field hockey players. Predominately senior the incorporation of the principles of players suffered chronic non-contact injuries prevention of injury in the overall training whereas, junior players suffered acute contact period.

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